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Course: CSA0414 - Operating System

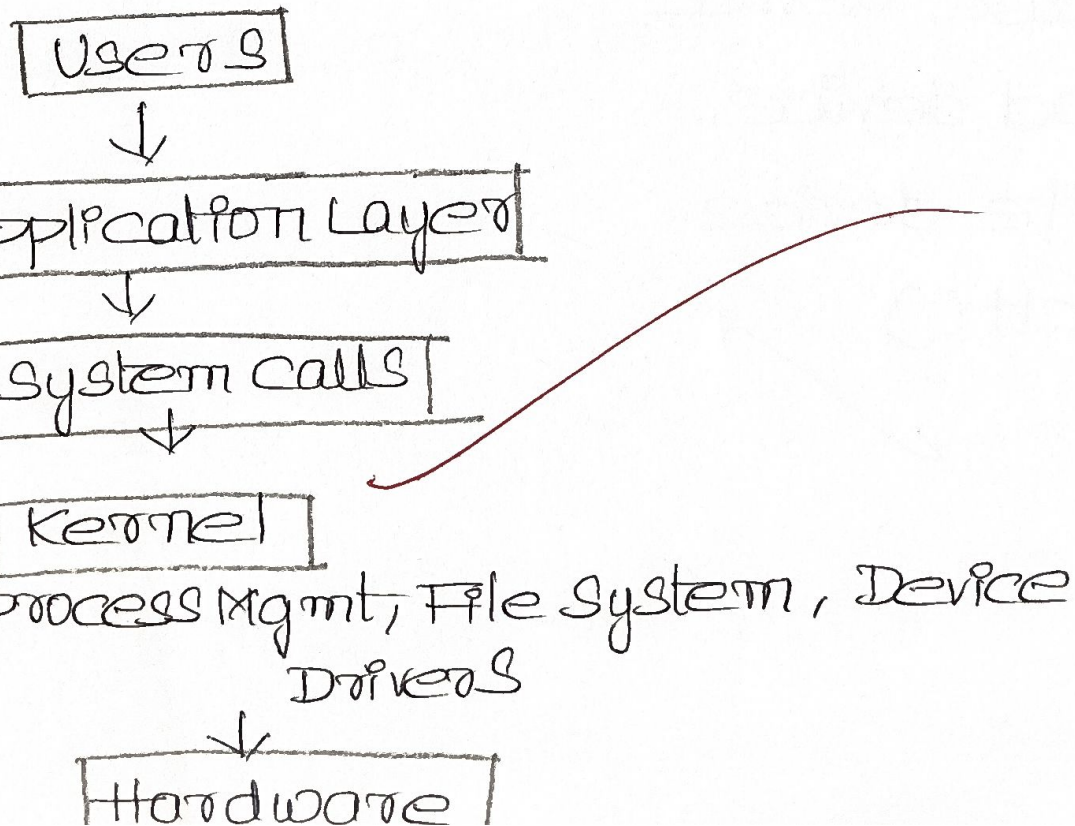
CO1 - Assessment Tool 1

SHORT ANSWER TEST

① A Modern System supporting multiple users and processes is best suited by:

1. Multi user operating system
   - Allows multiple users to access the system simultaneously
   Ex: Linux, UNIX
2. Time sharing operating system
   - Allocates CPU time slices to users and processes
   - provide quick respone and fair resource sharing

⑥ Structure of operating system:



(c) Sequence of system calls to create and execute a process

System call sequence

1. fork() - creates a child process
2. exec() - Loads and executes a new program
3. wait() - Parent waits for child completion
4. exit() - Terminate process.

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(2)

(a) File Management system calls

- create()
- open()
- read()
- write()
- close()
- delete()

Device Management system calls

- request device
- release device
- read device
- write device
- ioctl()

Layered structure for File operations

User Applications

↓

File system calls

↓

File system Layer

↓

Device Drivers

↓

Storage Hardware

2) C-Based system call sequence

```
#include <fcntl.h>
#include <unistd.h>

int main()
 {
int fd;
char buffer[100];
fd = open("data.txt", O_RDONLY);
read (fd, buffer, sizeof(buffer));
close(fd);
return 0;
}
```

- open() - opens the file
- read() - reads data
- close() - close the file descriptor

(3) (a) The suitable OS is Real Time Operating System (RTOS)

1. provides deterministic response time
2. Ensures critical tasks
3. Suitable for industrial and embedded system
4. High reliability and predictability

(b) Interaction b/w Hardware, Kernel and User Processes

User Processes

↓

Kernel

Scheduling

Memory Management

Synchronization

↓

Hardware

(c) System calls for process synchronization

1. fork():
   - creates cooperating processes
2. wait():
   - waits for another process
3. sem_wait():
   - Acquires semaphore

pusht();

- Releases semaphore

5. pthread_mutex_lock();
   - Locks critical section
6. pthread_mutex_unlock();
   - Unlocks critical section.

These calls prevent race conditions and ensure correct process coordination

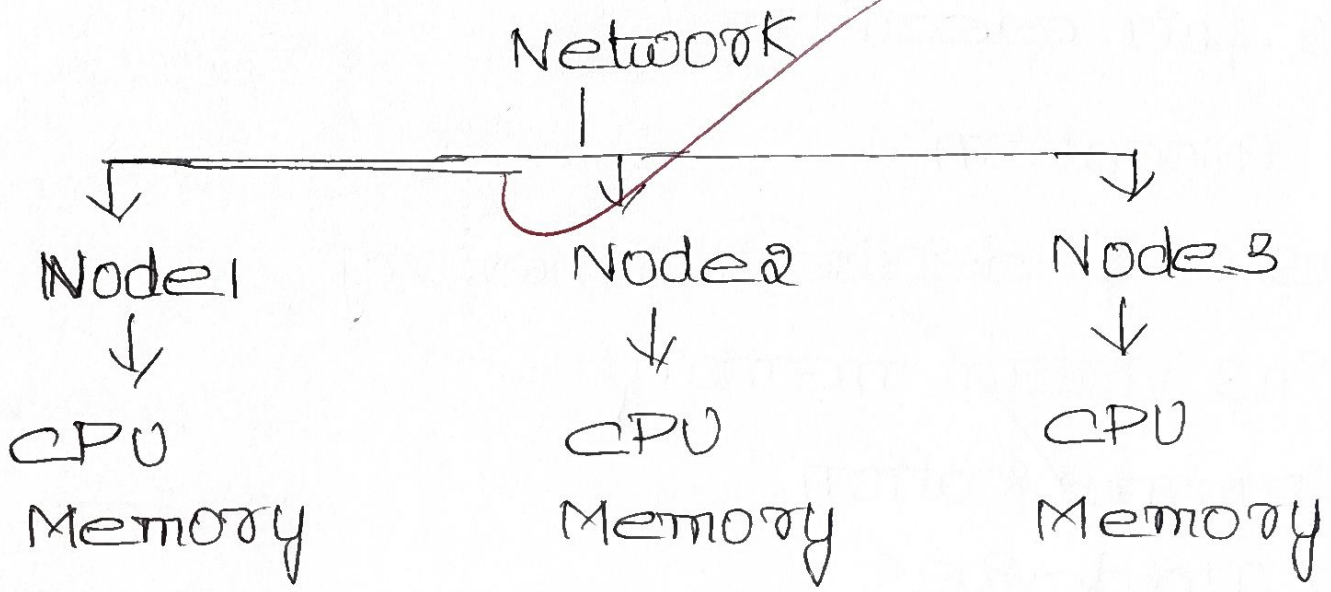
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(4)

(a) The system uses a Distributed OS

	Distributed OS	Centralized OS
1.	Multiple interconnected computers	Single computer system
2.	Resources shared across network	Resources shared locally
3.	Better scalability	Limited scalability

(b) Distributed OS architecture



All nodes communicate and share resources through the network

(c) IPC system call sequence

```
pipe(fd);
write(fd[1], "Hello", 5);
read(fd[0], buffer, 5);
close(fd[0]);
close(fd[1]);
```

Sequence:

1. pipe()
2. write()
3. read()
4. close()

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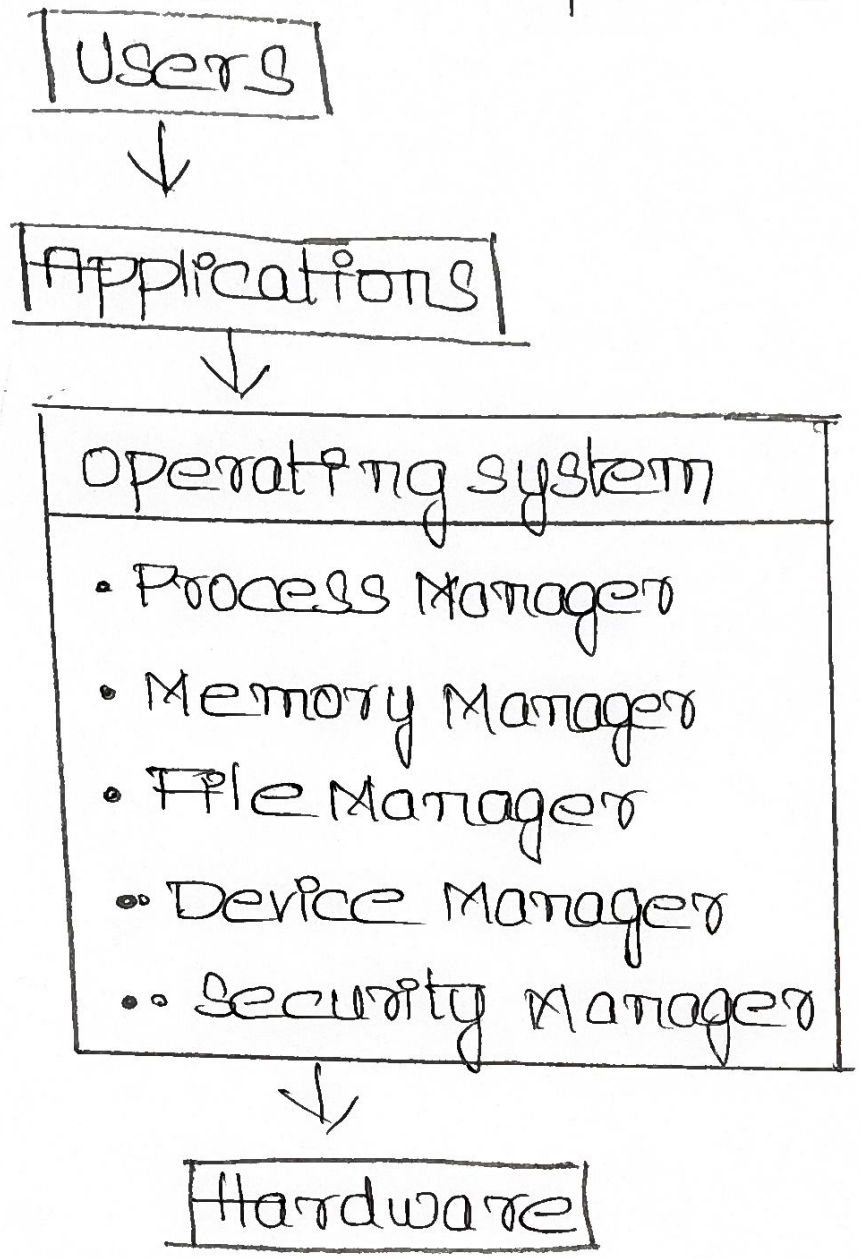
(5)

(a) Major functions of an operating system

1. Process scheduling
   - Allocates CPU to process
   - Ensures fair execution
2. Memory Allocation
   - Allocates and deallocates memory
   - Maintains virtual memory
3. Device communication
   - controls I/O devices
   - Uses device drivers.

these functions improve system efficiency.

Block Diagram of OS components



(c) System call examples

Memory Allocation

```
void *ptr;
ptr = malloc(100);
free(ptr)
```

System calls;

- brk()
- sbrk()
- mmap()

Device Management

System calls:

- open( )
- read( )
- write( )
- close( )